

Self-monitoring, pseudo-ductile, hybrid FRP reinforcement rods for concrete applications

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Abstract

The feasibility of hybrid -fiber-reinforced-polymer rods that demonstrate the important safety features of self-monitoring capability and pseudo-ductility is demonstrated. The rods are intended to be the basis of improved pultruded reinforcements for concrete or other civil applications where safety is of critical importance. The lowest elongation fiber in the seven rods investigated is carbon, which by virtue of its piezoresistivity allows the monitoring of deformation and fracture throughout an entire rod with simple electronic equipment. Resistance measurements obtained during quasi-static tests clearly reveal failure of the carbon fibers. Following this easily detected event, higher loads can be safely sustained by the remaining high-elongation fibers if the carbon tows are dispersed in the cross-section rather than concentrated in one location. © 2001 Elsevier Science Ltd. All rights reserved.

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1. Introduction

The types of fiber-reinforced-polymer (FRP) composites best suited for the reinforcement of concrete are those providing high strength, high stiffness, and environmental compatibility with concrete [1]. Most commercial FRP products are rod-like elements that are pultruded, shaped, and treated so that surface texture and undulations provide a mechanical interlock with concrete. Orienting the strong, stiff fibers in the longitudinal direction of the reinforcement enables the designers of the reinforcements to maximize the strength and stiffness of the composite material (with a calculated sacrifice of transverse properties). Maximizing longitudinal stiffness by a high degree of fiber alignment is done more for minimizing deflection of the structure rather than for maximizing strength since most FRP reinforcements have longitudinal tensile strengths well in excess of that of typical reinforcement steel. Stiffnesses of these FRPs, however, are usually in the range of 1/5–2/3 that of steel. Hence, maximum stiffness is of

high importance to designers of FRP reinforcements for concrete.

Another concern expressed by designers considering FRP reinforcements for concrete is safety. Since unidirectionally reinforced FRPs are needed to maximize stiffness and the high stiffness fibers that dominate the tensile mechanical response are essentially linear elastic to failure, there is potentially inadequate ductility in the reinforcement itself to satisfy existing requirements for civil structures. Overcoming this limitation of FRP materials could involve (a) implementation of design codes that allow the design of over-reinforced structures that fail by a graceful crushing of concrete rather than sudden rupture of the reinforcement; (b) devising more-ductile reinforcements that enable early detection of impending tensile failure; (c) devising more-sensitive means of inspecting reinforced concrete structures for signs of distress; or (d) some combination of these or other approaches. The approach of this investigation is a combination of items (b) and (c).

The objective of this investigation is to develop an FRP reinforcing material for concrete that is stiff, strong, and self-monitoring such that the overall deformation and damage of the embedded reinforcement can be easily measured by inspectors without separate sensors

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and elaborate electronics. In addition, a graceful failure mode, or pseudo-ductility, will be designed into the normally brittle composite material by proper selection of hybrid fiber reinforcement. Self-monitoring capability will be achieved using the piezoresistive effect in carbon fibers to monitor their strain and fracture. Clearly, FRP materials have a proven record for durability in harsh environments, as one realizes by consideration of the countless FRP boats afloat in salt-water for a half-century or so. If the safety issue on the minds of designers could be addressed by taking advantage of the nearly infinite tailorability of composites, significant benefits to society could be realized.

2. Literature review

The modulus of elasticity, E , of a longitudinally reinforced hybrid fiber composite is estimated by the so-called “rule of mixtures,” [Eq. (1)],

$$E = \sum_{i=1}^n E_i V_i \quad (1)$$

where E_i and V_i are the elastic moduli and volume fractions of the n constituents of the composite, including matrix and each type of fiber. A rough estimate of the longitudinal strength of a unidirectional composite consisting of several constituents subjected to a homogeneous strain field is based on the ultimate strain of the least ductile constituent. The formula for stress applied to the hybrid composite at failure of the least ductile fiber, $(\sigma)_{\text{ult}}$, is given by Eq. (2),

$$(\sigma)_{\text{ult}} = (\varepsilon_f)_{\text{ult}} E \quad (2)$$

where $(\varepsilon_f)_{\text{ult}}$ is the ultimate strain of the least ductile fiber and E the composite modulus from Eq. (1). Normally, $(\varepsilon_f)_{\text{ult}}$ is found by stressing impregnated, non-hybrid fiber strands to failure. In certain hybrid fiber composites, though, the recorded strain at which the least ductile fiber fails appears to differ from $(\varepsilon_f)_{\text{ult}}$. In the event of greater failure strain, a ‘positive hybrid effect’ is said to exist. Similarly, a ‘negative hybrid effect’ corresponds to a decreased in situ failure strain of a particular fiber when used as part of a hybrid composite. In some cases, failure of the least ductile constituent does not lead to immediate failure of the remaining constituent(s) if the remaining fibers can carry the redistributed load. This latter behavior leads to the pseudo-ductile stress/strain relationship and increased overall energy absorption that is considered attractive in many different loading situations, including creep and impact [2].

Hayashi [3] was one of the earliest investigators to report on the phenomenon of graceful failure in sandwich carbon and glass fiber hybrid composites. Bunsell and Harris [4] found that good bonding between the layers of sandwich hybrids improved the ‘smoothness’ of the stress-strain curve due to a more graceful transfer of load from carbon layers to glass layers, in comparison with unbonded sandwich hybrids. Several investigators in the 1970s believed that beneficial compressive residual stresses in carbon fibers were the cause of the positive hybrid effect often observed in the tensile-testing of carbon/glass hybrids, although this mechanism was later found to be insufficient to account for the effect [5]. More recently, the evidence suggests that the arrangement of the various fibers in hybrids governs the hybrid effect. Kretsis [6] reviewed the results of some 50 publications on multi-fiber hybrids in tension experiments and noted that the low elongation fiber demonstrates a positive hybrid effect while the high elongation fiber demonstrates a negative hybrid effect. Also, Kretsis noted that positive hybrid effects of up to 50% were most apparent in smaller proportions and concentrations of the low elongation fiber. Studies such as those by Rosettos [7], Jones and DiBenedetto [8], and Bader and Manders [9] attribute the positive hybrid effect in carbon fibers to the stress-blunting effect of many less-stiff, high-elongation fibers surrounding a few, dispersed carbon fibers. Hence, the apparent positive hybrid effect may be not be a consequence of the improvement in the ultimate strain of a given carbon fiber per se. Rather, the few carbon fibers dispersed in glass composites seem able to withstand higher strains, in the global sense, prior to large-scale failure than do carbon fibers in carbon-only composites. The ability of glass fibers to transfer load back into isolated broken carbon fibers at applied strains that would cause damage localization and catastrophic failure of carbon-only composites is the mechanism by which the positive hybrid effect is currently believed to occur in carbon/glass hybrids.

Other investigators have observed positive hybrid effects and pseudo-ductility in various unidirectional FRPs such as aramid/glass [10], carbon/polyethylene [11], and carbon/aramid/polypropylene [12]. Jones and DiBenedetto [8] investigated carbon/carbon, E-glass/E-glass (with strong and weak interfacial bonds), aramid/carbon, and E-glass/carbon. The conclusions from these studies that are pertinent for the present investigation are that the low-elongation fiber must be present in small amounts and must be well dispersed to obtain pseudo-ductility. The best levels of uniformity and/or strength of the fiber/matrix bond on macroscopic stress/strain behavior still seem to be subject to debate, however. Pseudo-ductility has been demonstrated by braiding [13,14] or filament winding [15] tows of different fibers.

The piezoresistivity of carbon fibers has been documented for many years [16]. The zero-frequency resistance change of longitudinally-reinforced carbon fiber composites is generally believed to arise from several sources: (a) geometry change caused by elastic deformation of fibers; (b) resistivity change caused by electrical changes within fibers; (c) resistivity change due to strain-dependent interfiber contact in a composite; and (d) fiber fracture [17,18]. A widely used measure of the elastic strain-sensing capability of a material is the gage factor, given by Eq. (3)

$$GF = \frac{\Delta R/R}{\varepsilon} \quad (3)$$

where ΔR is the change in zero-frequency resistance, R , and ε is the longitudinal strain. The resistance of a conductive material is dependent upon its resistivity, ρ , cross-sectional area, A , and length, L according to Eq. (4).

$$R = \frac{\rho L}{A} \quad (4)$$

By taking the total differential of Eq. (4), it is apparent that, as longitudinal stress is applied to a single carbon fiber and the fiber contracts laterally (as a result of a known Poisson's ratio of about 0.3), the resistance changes in proportion to elongation. However, it has been noted that the change in resistivity, ρ , with respect to strain can dominate the geometric resistance change in a single fiber. For instance, Crasto and Kim [17] noted that low-modulus carbon fibers (241–310 GPa) have increasing resistivity with tensile elongation whereas high modulus fibers (>480 GPa) have the opposite trend. When many carbon fibers are placed into a polymeric matrix, it is possible that the changing amount of inter-fiber electrical contact with respect to strain can also dominate the single-fiber geometric resistance change [18]. While gage factors of 1–2 are typically observed for stand-alone, low-modulus carbon fibers such as Hexcel AS4 and Torayca T300, gage factors for longitudinally-loaded unidirectional carbon FRP composites with these same fibers cover an order of magnitude larger range with either positive or negative sign [17–19].

Several researchers have demonstrated the use of carbon fibers for monitoring damage of carbon FRPs with quasi-static or cyclic tensile loads [19–21]. Researchers in Japan have demonstrated the use of hybrid fiber-reinforced FRP grids (orthogonal lattices) for health monitoring of concrete [22,23]. The process believed to dominate the irreversible increase in resistance of the least-ductile piezoresistive fibers in each load case, beyond a certain nominal fiber strain, is dispersed fiber fracture.

3. Experiments

Seven candidate hybrid FRP rods were designed, manufactured, and tested for their suitability in meeting the prescribed objectives. At this point in the technology development, the focus is on basic material design and evaluation. Bond with concrete should be investigated later. During tensile tests of stand-alone rods, load, strain, and direct current (DC) electrical resistance were recorded. The experimental methods, outlined below, have been fully detailed by Terosky [24] and Koehler [25].

3.1. Manufacture of hybrid rods

Smooth, 11-mm-diameter rods were pultruded at The Pennsylvania State University using varying combinations and dispersions of E-Glass (Owens Corning 366-AI-113), high tenacity poly (vinyl alcohol), referred to as PVA (Kuraray Vynolon® Type 7901), aramid (DuPont Kevlar® 49) and carbon (Hexcel AS4C-GP-12K) fibers in polymer resin matrices. The two types of resins used in this investigation were unsaturated polyester (Reichhold PolyLite® 31-031) and vinylester (Ashland Hetron D-1222). Other ingredients added to the resins included inert filler, catalyst, and release agent. The fibers were chosen based on their reasonable cost and high tensile strength and modulus. A good spread of ultimate fiber elongations was desired for obtaining pseudo-ductility. In addition, the lowest-elongation fiber (carbon) needed to be piezoresistive in order to obtain self-monitoring capability. Polyester and vinylester resins are ideal for pultrusion because of their low cost, low viscosity, fast cure, and high chemical shrinkage. No effort was made to optimize the matrix or fiber surface treatments in this research. Some pertinent properties of the constituents of the rods are listed in Table 1. Fiber-volume fractions and carbon fiber placement patterns of all seven types of rods are listed in Table 2. The seven material designs were selected to interrogate the theories of self-monitoring and pseudo-ductile hybrid FRPs discussed in the literature review, within the confines of practical production capabilities, and to provide guidance for future material optimization efforts.

In the pultrusion process line, dry tows of fiber were continuously pulled from spools, impregnated with resin, consolidated by drawing through a series of progressively smaller dies, and cured in a 1-m-long heated die, as detailed by Bakis et al. [26]. Peak die temperature was selected to be no greater than approximately 160°C to avoid the recommended 180°C short-term limit for PVA fibers [27]. Pultrusion rates were typically 22–30 cm/min.

Prior to entering the heated die, the wetted tows were oriented to their preferred positions using guides. Fiber placement in the mold was such that carbon tows were

either concentrated in the center of the rod as in Type A and B rods, dispersed throughout the rod as in Type D, or dispersed over the surface of the rod as in Types C, E, F, and G (Fig. 1). The degree of dispersion of the

fibers was limited by the number of fiber spools that could be fed and tensioned at once (32) and the cross-section of the individual tows easily available at the time. For instance, one 12-K tow of carbon results in a relatively small area fraction of carbon fibers in one location (about 0.5%, as in Type C rods), but many such small tows could not be widely dispersed through the cross-section since more than 32 spools of fiber would be needed to fill the mold. The compromise approach was to re-spool three or four 12-K tows of carbon onto one spool and place each of them in separate groups in the cross section. Such was the case in Types E, F and G rods, where four groups of carbon tows were placed at roughly equal spacings around the perimeter of the rod. The resulting 6% volume fraction of carbon fiber in these cases was considered adequate for modulus enhancement and yet not excessive to the extent that pseudo-ductile behavior was prevented.

Table 1
Typical fiber and matrix properties

Material	Elongation (%)	Young's modulus (GPa)	Tensile strength (MPa)	Filament diameter (μm)
Hercules AS4C carbon	1.6 ^a	236 ^a	3860 ^a	7
Kevlar® 49 aramid	2.4 ^a	112 ^a	3000 ^a	12
E-glass	3.6 ^a	73 ^a	2580 ^a	14
Vynlon® PVA	4.9 ^a	45 ^a	2010 ^a	14
Unsat. polyester	3.3	3.4	75	Not applicable
Vinylester	4.0	3.6	80	Not applicable

^a Value based on impregnated strand test.

Table 2
Fiber composition of pultruded specimens

Group	A	B	C	D	E	F	G
Fiber volume fraction ^a	35% G 13% C	24% G 23% P 9% C	60% G 0.5% C	35% G 13% C	27% P 17% A 6% C	42% G 6% C	27% P 17% A 6% C
Resin ^b	PE	PE	PE	PE	PE	PE	VE
Carbon-fiber placement	In core	In core	On surface	Dispersed	On surface	On surface	On surface

^a G = glass; C = carbon; A = aramid; P = PVA.

^b PE = polyester; VE = vinylester.

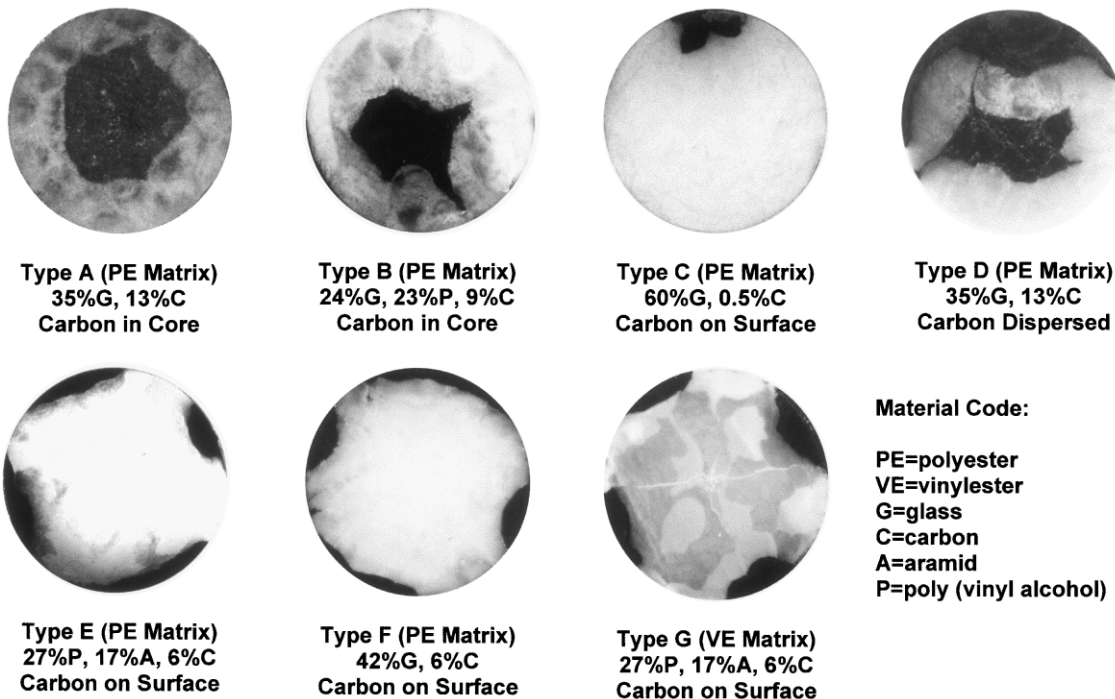


Fig. 1. Photographs of cross-sections of hybrid FRP rods (each cross-section is 11 mm in diameter).

Type A and B rods differed in the amount of carbon fiber in the core and in the type of high elongation fibers used. Type C had the least amount of carbon of all and only glass as the high-elongation fiber. Type D had the same composition as type A, but the carbon fibers were more dispersed in the former. Types E and G contained an intermediate amount of dispersed carbon and no glass, differing only in resin type. Type F resembled types E and G except that the aramid and PVA fibers of the latter two were replaced entirely with glass. In all types, the higher-strain fibers were evenly dispersed, tow-by-tow, throughout the cross-section.

3.2. Specimen preparation

Multiple specimens of each type were cut to approximately 1-m lengths with a water-cooled diamond saw. Conical depressions were machined into both ends of the rod and filled with a conductive epoxy to improve electrical contact with all the conductive fibers in the cross-section of the rod. Conical, epoxy-potted, filament-wound glass/epoxy FRP anchors were next fixed near the ends of the rods and served to transfer load smoothly into the rods and also insulate the rods from the metallic load frame. Electrical connections were accomplished by bonding aluminum foil to the cut ends of the rod protruding from the anchor with conductive epoxy and wrapping a copper lead-wire around the foil. A two-resistor voltage divider circuit was used to determine the DC resistance of the rod being tensile tested (Fig. 2). The dummy resistor of the divider circuit was either a fixed ceramic resistor or a nominally identical pultruded rod.

3.3. Testing

Three specimens of each group were tensile tested in a screw-driven universal testing machine. In the quasi-static strength tests, the displacement rate was chosen

such that specimens failed in approximately 2–3 min. Longitudinal strain was measured by either a 25-mm extensometer or a bonded foil resistance strain gage. Both methods were not reliable past the first failure event due to excessive recoil. The number of replicate mechanical tests done per rod is shown in Table 3. A sustained load test and a cyclic load test were carried out to determine the effect of loading history on the load-resistance behavior of type F rods.

Electrical resistance of the rod being tested was computed using the voltage divider method [28]. The change in voltage across the specimen (Fig. 2) is given by

$$\Delta V_{\text{specimen}} = \frac{r}{(1+r)^2} \left(\frac{\Delta R_{\text{specimen}}}{R_{\text{specimen}}} - \frac{\Delta R_{\text{dummy}}}{R_{\text{dummy}}} \right) (1-\eta)V_{\text{source}} \quad (5)$$

where Δ indicates a change in a quantity, r is the resistance ratio $R_{\text{dummy}}/R_{\text{specimen}}$, and η is a non-linear term given by Eq. (6).

Table 3
Mechanical and electrical results

Group		Modulus (GPa)	1st failure strain (%)	First strength (MPa)	Gage factor
A	No. tests	4	4	4	2
	Average	57.2	1.2	706	2.9
	CV	0.9	14.5	16.2	7.4
	Predicted	57	1.6	855	–
B	No. tests	4	4	4	3
	Average	42.3	1.7	596	2.3
	CV	10.8	19.9	4.6	9.9
	Predicted	48	1.6	720	–
C	No. tests	3	3	3	2
	Average	42.3	1.4	609	1.4
	CV	2.8	5.7	0.9	–
	Predicted	45	1.6	675	–
D	No. tests	2	2	3	1
	Average	53.9	1.1	569	3.8
	CV	3.4	3.3	2.5	–
	Predicted	57	1.6	855	–
E	No. tests	3	3	3	2
	Average	47.2	1.1	485	2.1
	CV	3.1	16.6	5.9	–
	Predicted	42	1.6	630	–
F	No. tests	4	4	4	3
	Average	43.4	1.2	543	2.7
	CV	4.6	15	11	7.8
	Predicted	44	1.6	660	–
G	No. tests	4	4	4	3
	Average	43.9	1.3	574	1.3
	CV	0.1	10	6	4.3
	Predicted	42	1.6	630	–

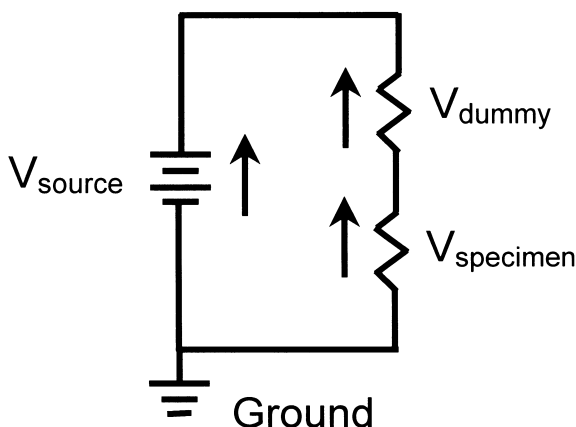


Fig. 2. Schematic diagram of voltage divider circuit.

$$\eta = 1 - \frac{1}{1 + \left[\frac{1}{(1+r)} \right] \left[\frac{\Delta R_{\text{specimen}}}{R_{\text{specimen}}} + r \left(\frac{\Delta R_{\text{dummy}}}{R_{\text{dummy}}} \right) \right]} \quad (6)$$

For resistance changes less than 5–10%, the nonlinear term is normally neglected. For monitoring of resistance in hybrid rods after the onset of carbon fiber fracture, the nonlinearity should be included in Eq. (5) since a several-order-of-magnitude change in resistance can occur. The thermal-strain-induced resistance changes in an active specimen and an identical, unloaded rod serving as the dummy resistor in Eq. (5) cancel when the non-linear term is negligible, thus achieving temperature compensation in such cases.

Proceeding under the presumption that temperature changes will be small in the laboratory, one can easily solve the nonlinear form of Eq. (5) for the mechanical part of $\Delta R_{\text{specimen}}$ as follows:

$$\Delta R_{\text{specimen}} = \frac{-\Delta V_{\text{specimen}}(R_{\text{specimen}} + 2rR_{\text{specimen}} + r^2R_{\text{specimen}})}{\Delta V_{\text{specimen}} + \Delta V_{\text{specimen}}r - rV_{\text{source}}} \quad (7)$$

Resistance change computed with Eq. (7) during tensile testing was used with Eq. (3) to determine the gage factor in electrically tested rods. The sensitivity of the resistance measurements varied from 0.01 and 0.04 Ω , depending on the exact bridge completion resistor and data acquisition system used. The number of replicates of electrical tests for each type of rod is given in Table 3. Electrical and mechanical data were collected digitally at a rate of 4 Hz.

4. Results

4.1. Manufacturability

Manufacturing of the rods went as planned except for some difficulties in controlling the cross-sectional placement of dispersed carbon tows (Type D, in particular, see Fig. 1). These difficulties arose due to the tendency of isolated carbon tows to float in the consolidation portion of the pultrusion line. This behavior did not occur when the carbon tows were concentrated as in rod types A and B. To circumvent this problem, subsequently manufactured rods having dispersed carbon tows (types E, F, and G) were made by introducing the carbon fibers to the bulk of the cross section just prior to entering the heated mold. This effectively prevented the carbon tows from migrating upwards, although it constrained the placement of the carbon tows to the surface of the rod. This limitation could easily be overcome in

the future by using tows manufactured with prescribed hybrid fiber compositions.

4.2. Mechanical properties

The results of the quasi-static tensile tests are summarized in Table 3. The initial moduli (before first fracture) were generally within $\pm 6\%$ of predictions by Eq. (1) when using just glass and carbon and within $\pm 12\%$ of predictions in the other cases. The first failure events were noticeable as jogs in the load versus time data, as shown in representative cases in Figs. 3–8. Based on visual observations and the proximity of the first failure strains to the ultimate strain of impregnated carbon

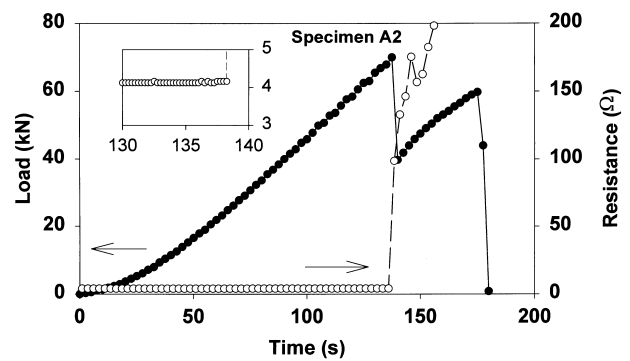


Fig. 3. Load and resistance to failure for typical type A specimen.

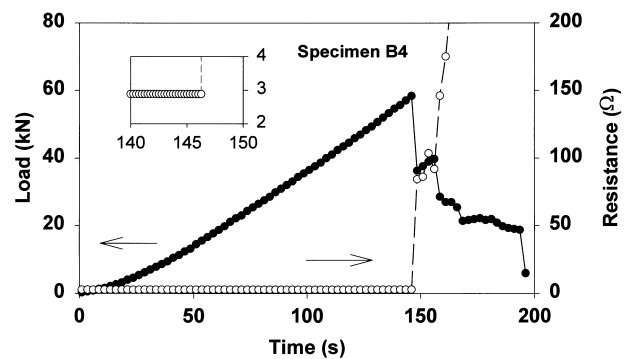


Fig. 4. Load and resistance to failure for typical type B specimen.

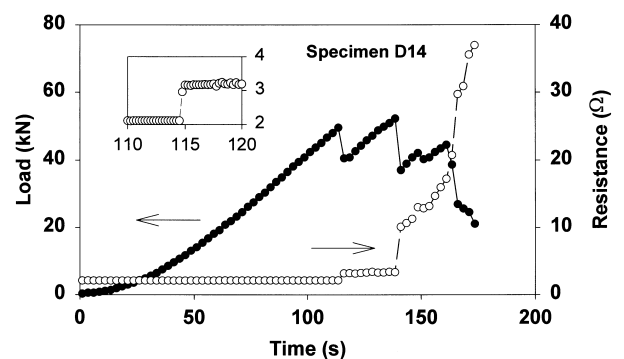


Fig. 5. Load and resistance to failure for typical type D specimen.

strands (1.6%), it is concluded that first failure occurred in carbon, as planned. First failure strains ranged from 31% below the carbon strand value in type D rods to 6% above the strand value in type B rods. Given the small sample size and the knowledge that ultimate strains of normal coupon-sized carbon composites are generally 15–20% below impregnated strand ultimates due to a variety of strength-reducing effects such as size, gripping method, and fiber damage or misalignment during processing, the first failure strains obtained in this investigation are within expectations. There was no clear trend in first failure strain with respect to dispersion or concentration of carbon fibers or type of high elongation fibers.

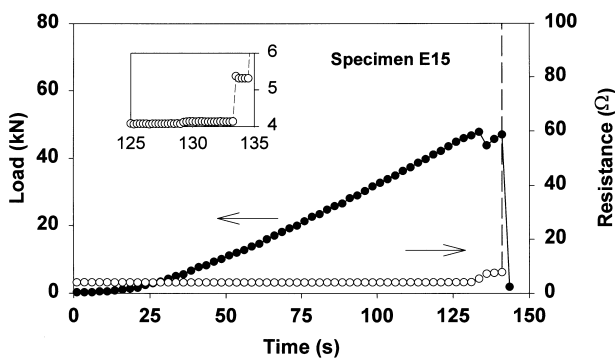


Fig. 6. Load and resistance to failure for typical Type E specimen.

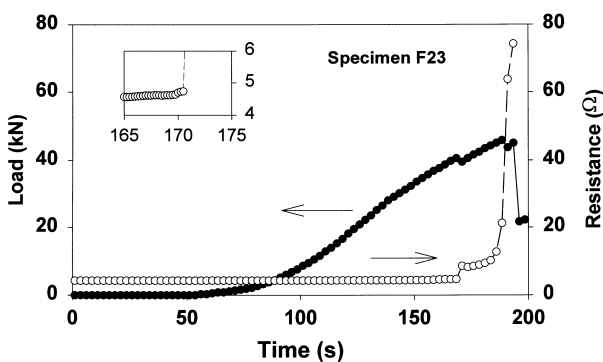


Fig. 7. Load and resistance to failure for typical Type F specimen.

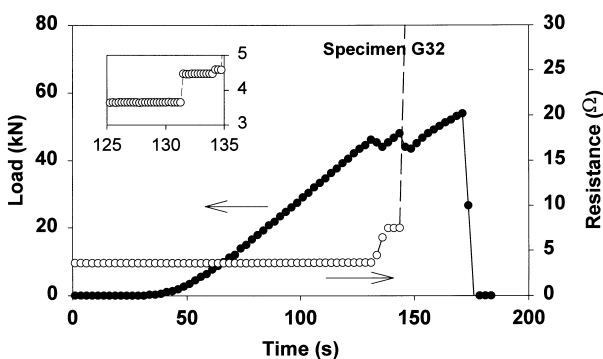


Fig. 8. Load and resistance to failure for typical Type G specimen.

All of the rods had first tensile strengths well above the usual value of 410 MPa for steel reinforcing bars for concrete. As expected, the moduli were significantly lower than steel (210 GPa), especially in those rods having fewer carbon fibers. Nearly all of the rods had some degree of load carrying capacity following first failure. This was most prominent in type D, F, and G rods, where at least one subsequent load peak exceeded the first load peak in most replicate experiments. The rods having the most numerous and highest load peaks following first failure were type F and G. Most of these had two or three subsequent load peaks greater than or equal to the first failure load. The most extreme example was five subsequent load peaks greater than the first failure load in a type F rod. Overall, more pseudo-ductility of this sort was observed in specimens having a greater dispersion of carbon fibers in the cross-section, regardless of the types of high-elongation fibers present and the amount carbon fiber present. Type A and B rods had the highest volume fraction and concentration of carbon fiber, the highest modulus, and the least load-carrying capability following first failure. Although the results from type C rods are not included here for brevity, it is worthwhile to note that these rods had relatively little pseudo-ductility.

The glass, carbon, and aramid fibers demonstrated good pull-out in failed specimens, while the PVA fibers were noted to break on a common plane perpendicular to the loading direction. Such behavior is typical of PVA fibers due to their extremely strong bond with many polymer resins [29] and suggests that this type of fiber may not be suitable for the objective of pseudo-ductility in hybrid rods. In cases where the carbon fibers were on the outside of the rod, the carbon fibers tended to separate from the remainder of the rod after first failure. This behavior may be an advantage for minimizing stress concentrations on the high elongation fibers, but it is not known what effect it would have on the bond performance of a reinforcement rod in concrete.

4.3. Electrical properties

Typical initial resistances of the rods were between 3 and 25 Ω . As expected, rods with less carbon fiber (type C, for instance) had higher initial resistances. The variation of resistance with strain under monotonic loading, as quantified by the gage factor [Eq. (3)], was rather independent of rod type and ranged from 1 to 4 (Table 3). Considering all specimens electrically tested, the average gage factor was 2.2. This value is in accordance with previous results cited in the literature for AS4 carbon fiber.

In a type F rod held at approximately 50% of ultimate elongation for 24 h, the resistance gradually increased by 0.1 Ω , or 3% of the initial value. Over this same period of time, the room temperature decreased by

2°C and the load decreased by about 10%. This magnitude of resistance change is comparable to that typically observed to the point of first failure in quasi-static tests. In a type F specimen cyclically loaded and unloaded six times to 50% of first failure strain, no change in the gage factor was observed for each reloading. A barely measurable 0.01 Ω increase in resistance (less than 0.2% of initial resistance) was noted between the third and fourth cycles, however. For reference, the 50%-of-ultimate strain level is commonly regarded as below the 10^6 -cycle fatigue endurance limit of longitudinally-loaded, unidirectional, carbon-fiber composites.

Upward jumps in the resistance versus time data (Figs. 3–8) corresponded well with the jogs in the load data. It is not possible to discern the exact mode of subcritical failure by monitoring electrical resistance in this manner, but it is certainly quite evident when some type of damage has occurred in the rod. Since several of the hybrid rod types sustained higher loads subsequent to the first large jump in resistance, the capability of hybrid rods in providing a useful warning of catastrophic failure is thus demonstrated. The inset graphs of Figs. 3–8 provide a detailed record of the resistance in 10-s periods of time near first failure. In most cases, the jump in resistance at first failure occurred within less than 0.25 s. Considering the overall plot of the resistance versus strain in several specimens, however, the slope gradually and monotonically deviated from linearity at strains exceeding 0.5–0.8% — well before the more obvious mechanical and electrical evidence of first failure. Further investigation is needed to determine if such slight deviations are reversible (i.e. related to gage factor) or permanent (i.e. related to structural changes or local failures of individual carbon fibers).

5. Conclusions

Hybridization of unidirectionally reinforced composites using low-elongation piezoresistive fibers and high elongation electrically inert fibers is a viable method of obtaining pseudo-ductile tensile behavior with a built-in capability to monitor full-field strain and damage. Pultruded rods with low concentrations of dispersed carbon tows were better able to sustain further loading following first failure of the carbon tows. In such cases, much advance warning of an impending ultimate load condition was obtained using simple electronic circuitry. Type D, F and G rods, each having between 6 and 13% by volume carbon fibers dispersed in the cross-section, displayed the best combination of pseudo-ductility and early-warning of catastrophic failure by resistance measurements made across opposite ends of the specimens.

Typical values of the so-called strain gage factor for the carbon fibers ranged from 1 to 4. Gage factors were not significantly altered by several load cycles of a rod

to 50% of the ultimate tensile strain of carbon, although the rate of increase in resistance with respect to strain was noted to slightly accelerate at strains greater than about 50% of ultimate in quasi-static tensile tests to failure. Resistance changes that occurred under a 24-h low-level sustained loading were comparable to those that occurred in quasi-static loading to first failure. Hence, it is possible that, with further development, the proposed self-monitoring, pseudo-ductile hybrid FRP rods could be an effective means of reinforcing and monitoring concrete structures.

6. Recommendations

The smooth rods used in this investigation should be further developed with surface undulations and/or coatings to enhance bond with concrete. Individual tows should be hybridized to minimize the floatation problem in pultrusion and possibly enhance the positive hybrid effect in the low elongation fibers by improved the mixing of fiber types. While the response of hybrid rods to monotonic loading to failure is of relevance to the problem of localized high strain on a reinforcement rod in an overloaded concrete structure, the more likely loading scenario of interest is that of creep rupture. Tests of this type should be carried out to determine the early-warning effectiveness and pseudo-ductility of hybrid fiber rods embedded in concrete beams and slabs.

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